

SURGICAL ASPECTS OF INFLAMMATORY BOWEL DISEASE

Overview

- **IBD = idiopathic intestinal inflammation.**
- **Ulcerative colitis:** rectum + colon; **mucosa + submucosa only**; **rectal bleeding, tenesmus, mucus** (covers stool); **90% confined to rectum** but may extend; bloody diarrhoea + urgency; severe → anaemia, hypoproteinaemia, electrolyte disturbance.
- **Crohn's disease:** **chronic full-thickness (transmural)**, any part **mouth** → **anus**, **discontinuous (skip lesions)**.
- **Indeterminate colitis (10–15%):** features of both; treatment differs.

Clinical & severity

- Depends on diagnosis, location, extent. Proctitis = no systemic upset. Colitis = bloody diarrhoea + urgency. Children → impaired growth.
- **Severity classification (UC):**
- **Mild:** <4 stools/day, no toxicity.
- **Moderate:** >4 stools, mild anaemia, ↑ESR/CRP.
- **Severe:** >6 bloody stools, **fever, tachycardia, anaemia, hypoalbuminaemia (ominous)**.
- **Fulminant:** >10 movements, continuous bleeding, distension → **toxic megacolon (emergency surgery to avoid perforation)**.

Extra-intestinal manifestations

- **Arthritis (~15%, large joint), sacroiliitis, ankylosing spondylitis, sclerosing cholangitis, erythema nodosum, pyoderma gangrenosum, uveitis, episcleritis.**

Cancer risk (UC)

- Increases with duration: **~1% at 10 yrs → 10–15% at 20 yrs → 20% at 30 yrs. Pancolitis >10 yrs → endoscopic screening for silent dysplasia.**

Surgery for UC

- **Lifetime colectomy risk ~20%.**
- **Indications:** severe/fulminant failing medical therapy, chronic disease (anaemia/urgency/tenesmus), **steroid-dependent**, drug intolerance, growth retardation, **neoplastic change (dysplasia/carcinoma)**, sclerosing cholangitis, extraintestinal manifestations, haemorrhage/stenosis.
- **Emergency: subtotal colectomy + end-ileostomy** (safest; rectosigmoid remnant as mucous fistula).
- **Elective:** subtotal colectomy + ileostomy; **proctocolectomy + permanent end-ileostomy; restorative proctocolectomy + ileal pouch-anal anastomosis (IPAA);** subtotal colectomy + ileorectal anastomosis.
- **IPAA:** usually 2-stage with covering loop ileostomy. Complications: pelvic infection (anastomotic leak), SBO (10–15%), pouch-vaginal fistula, **pouchitis** (→ metronidazole/ciprofloxacin + probiotics).
- **VTE prophylaxis:** IBD = **3× VTE risk** (higher in UC); post-op LMWH up to 28 days.

Surgery for Crohn's disease

- **Surgery does NOT cure** — manages complications. **Preserve healthy gut (avoid short bowel syndrome); minimal resection.**
- **Ileocaecal resection** (terminal ileal disease) — primary anastomosis (ileostomy if unwell/infected/depleted).
- Segmental resection (strictures); colectomy + ileorectal anastomosis (rectal sparing); subtotal colectomy + ileostomy; temporary loop ileostomy; **panproctocolectomy + permanent ileostomy** (severe); **stricturoplasty**.
- **Perianal Crohn's**: commonest = **perianal abscess** → fistula. Surgery controls infection; **seton** (allows biologic initiation); **infliximab/adalimumab + seton**; LIFT, advancement flaps, fibrin glue.

High-yield one-liners

- **UC = mucosal, continuous from rectum; Crohn's = transmural, skip lesions, mouth→anus.**
- **Severity: mild <4 / moderate >4 / severe >6 bloody stools; fulminant >10 → toxic megacolon (emergency surgery).**
- **UC cancer risk ↑ with duration (20% at 30yrs); pancolitis >10yrs → screening.**
- **UC surgery curative (colectomy); emergency = subtotal colectomy + end-ileostomy; elective = IPAA.**
- **Crohn's surgery NOT curative (complications only); minimal resection; perianal = seton + biologics.**
- **IBD = 3× VTE risk → LMWH 28 days; pouchitis → metronidazole.**

SURGICAL ASPECTS OF IBD — Revision Table

Bold = high-yield. Dr Farman Hassan.

UC vs CROHN'S & SEVERITY

Feature	Ulcerative colitis	Crohn's
Extent	Mucosa + submucosa; continuous from rectum	Transmural; skip lesions; mouth→anus
Symptoms	Rectal bleeding, tenesmus, mucus	Variable; perianal disease
Severity (UC)	Mild <4 / moderate >4 / severe >6 bloody stools; fulminant >10 → toxic megacolon	—
Extra-intestinal	Arthritis, sacroiliitis, sclerosing cholangitis, erythema nodosum, pyoderma gangrenosum, uveitis	Same
Cancer risk	↑with duration: 1% (10yr) → 20% (30yr); pancolitis >10yr → screening	Lower

SURGERY

Topic	Key points
UC surgery	Curative (colectomy); lifetime risk ~20%; indications: fulminant, steroid-dependent, dysplasia/cancer, sclerosing cholangitis
UC emergency	Subtotal colectomy + end-ileostomy
UC elective	Proctocolectomy + ileostomy; IPAA (ileal pouch-anal anastomosis)
IPAA complications	Pelvic infection (leak), SBO 10–15%, pouchitis → metronidazole/cipro + probiotics
Crohn's surgery	NOT curative — complications only; minimal resection (avoid short bowel); ileocaecal resection; stricturoplasty
Perianal Crohn's	Abscess → fistula; seton + infliximab/adalimumab; LIFT, advancement flap
VTE	IBD = 3× risk; LMWH up to 28 days post-op